

Raghav Kanda

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Education

University of Toronto

Toronto, Canada

Sept. 2021 – Apr. 2026

Bachelor of Science (HBSec) | Computer Science | Applied Statistics | Mathematics

Skills

Java, Python, C, C++, Go, Kotlin, SQL, Bash, Spring Boot, FastAPI, Django, Linux, Docker, Git, MySQL, REST APIs, Information Systems, Network Security, Cryptography, Systems Programming, Object-Oriented Programming, MVC, MVVM, Software Architecture, Algorithms, Machine Learning, AWS, Azure, GCP

Experience

Lead Android Developer

Mopply Inc

Toronto, Canada

Jan. 2025 – Present

- Architected production Android apps (Kotlin, Java, MVVM) and refactored legacy client/backend systems into reusable modules, improving maintainability and architecture by 30%.
- Integrated backend REST APIs and ensured secure data flow by implementing local data storage with advanced encryption and authentication protocols.
- Optimized system performance and memory overhead by 20% via efficient data caching and robust application management.

Software Developer

Freelance

Remote

May 2023 – Present

- Designed and developed secure custom software applications, backend information systems, automation tools, and data pipelines. (Python, Java, SQL, REST APIs)
- Built modular, scalable software solutions applying clean architecture principles, secure API integrations, and robust system deployment practices.
- Collaborated with clients to analyze system requirements, patch vulnerabilities, and deliver scalable, high-performance software implementations across 10+ projects.

Projects

Cybersecurity Exploits & Systems Security

C, Python, Linux, Systems Programming, Network Security

- Analyzed concurrency vulnerabilities and developed controlled buffer overflow exploits in C to mitigate memory corruption.
- Built a Python simulation of the DH Key Exchange protocol to demonstrate secure cryptographic network communications.

Algebraic Cryptography Library

Python, Cryptography, Algorithms, Data Security

- Implemented RSA, ElGamal, Affine, and Elliptic Curve cryptographic algorithms for secure data transmission.
- Developed reusable utilities for key generation, digital signatures, primality testing, and secure encryption/decryption.

Java Tetris Game with AI Autopilot

Java, JavaFX, MVC Architecture, Object-Oriented Programming

- Built a Java-based Tetris game using JavaFX and MVC architecture.
- Developed an AI Autopilot with heuristic evaluation and persistent save/load functionality using secure Java serialization.

Abstract Syntax Tree Evaluator

Java, Compiler Design, Visitor Pattern

- Built an Abstract Syntax Tree evaluator for parsing and computing nested mathematical expressions.
- Applied the Visitor pattern and implemented robust custom exception handling to prevent runtime execution failures.

Java Boggle Game

Java, JavaFX, MVC, Design Patterns

- Developed an interactive JavaFX Boggle game with dynamic board generation and dictionary-based validation.
- Applied MVC with the Memento and Command design patterns for modular architecture and game-state persistence.